High-Accuracy Crater Detection for Complex Planetary Surfaces

Miao Zhuo, Yiran Wang, Xiaoran Zhang, and Zeyu Chen

***Abstract*— Impact craters play a crucial role in planetary science, as they serve as windows into the geological evolution of planetary surfaces and are key considerations in selecting landing sites. However, most existing crater detection algorithms are either closed source or difficult to use due to complex configurations, and many are limited to specific celestial bodies, performing well only on simple, flat surfaces. To address these limitations, we present the "automatic crater detection (ACD) tool," a user-friendly software solution built on deep learning algorithm that was selected through experimental parameter comparisons. By incorporating active learning techniques, the tool continually improves through the inclusion of samples from varying landforms. This allows for accurate crater detection on complex and diverse planetary surfaces, with a minimum detectable crater diameter of 10 pixels. Tested on standard image datasets from the Moon, Mars, and Mercury, the method achieved high accuracy, achieving $F1$-scores of 92.8%, 91.7%, and 92.2% for the Moon, Mars, and Mercury, respectively. The ACD tool is wrapped with an interactive user interface that ensures easy utilization, allowing users to generate ready-to-use output data with a simple click. The introduction of ACD tool can empower scientists with a straightforward yet powerful tool for conducting crater research more efficiently, supporting tasks such as surface age estimation and landing site safety analysis. The software can be downloaded via the following link: https://www.scidb.cn/s/yemuYb.**

***Index Terms*— Crater detection, deep learning, Mars, Mercury, Moon.**

I. Introduction

IMPACT craters, as lasting imprints of celestial collisions, serve as windows into the geological evolution of planetary surfaces. Their identification is crucial for unraveling the surface ages [1], [2], [3], [4] and understanding the processes that have shaped planetary bodies over time [5], [6], [7], [8]. Moreover, the distribution of impact craters is a critical consideration in the selection of landing sites for spacecraft and rovers [9], [10], [11], [12]. With the advancements in deep space remote sensing technology, we have gained access to increasing high-resolution data. Given the enormous amount of data generated by high-resolution imaging, manual mapping of individual craters across high-resolution images is highly impractical. As a result, there has been a significant rise in the development and deployment of automated crater detection algorithms.

The identification and analysis of impact craters have traditionally relied on rule-based approaches, such as the Hough transform [13], template matching [14], pattern recognition [15], or combinations thereof. These approaches are highly dependent on the specific design of the algorithms, often necessitating extensive manual intervention. For example, the parameters for thresholding and edge detection must be meticulously adjusted for each new dataset, making the process labor-intensive and time-consuming. Moreover, these rule-based methods often struggle with variations in crater appearances due to differences in lighting, resolution, and surface conditions in the images, resulting in inconsistent and sometimes inaccurate outcomes.

A second family of algorithms relies on classic machine learning approaches, such as modified boosting [16], [17], AdaBoost [18], genetic algorithms [19], random forests [20], and support vector machines (SVMs) [21]. These methods rely on manually extracted features, limiting their automation and flexibility. Their performance also depends heavily on the quality and diversity of the training data and demands significant computational power for feature extraction and training. Due to these constraints, there has been a shift toward deep learning techniques, which automatically learn and extract features from raw data, reducing the need for manual intervention and improving performance, making them more suitable for the complex nature of planetary crater detection.

Deep learning techniques, particularly convolutional neural networks (CNNs), have revolutionized crater detection by improving accuracy and reducing the need for manual feature selection. For Moon crater detection, Yang et al. [22] introduced a two-stage method using R-FCN [23] and transfer learning, achieving recall rates of 94.71% with Chang'E-1 (CE-1) data and 93.35% with Chang'E-2 (CE-2) data. Building on the Faster R-CNN [24] and Mask R-CNN [25], Zang et al. [26] proposed Crater R-CNN, which addressed the challenges of detecting small craters and handling incomplete data annotations. Their approach achieved a precision of 90.50% and a recall of 63.50% on CE-2 Digital Orthophoto Map (DOM) data. Yang et al. [27] introduced

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domain adaptation techniques using automatically labeled synthetic data samples, effectively detecting unannotated real data with a precision of 81.00% and recall of 82.33% on Lunar Reconnaissance Orbiter (LRO) [28] Narrow Angle Camera (NAC) data. Fairweather et al. [29] adapted YOLOv3 [30] for LRO NAC images, detecting craters as small as 20 m with a recall of 93.00%. Later, Fairweather et al. [31] further improved crater detection with YOLOv5 [32] on Kaguya Terrain Camera (TC) [33] data, achieving a precision of 94.00% and recall of 98.00% for craters larger than 100 m in diameter. This new model also corrected the 15% overestimation in crater size that was observed in the YOLOv3 NAC model [29]. La Grassa et al. [34] introduced super-resolution techniques to significantly enhance the spatial resolution of planetary surface images, thereby improving crater detection performance, achieving a precision of 89.90% and recall of 87.20% on LROC wide angle camera (WAC) data. Turning to Mars crater detection, Benedix et al. [35] applied a YOLOv3-based crater detection algorithm to Thermal Emission Imaging System (THEMIS) infrared (IR) data [36], achieving a recall of 91.00%. Hsu et al. [37] improved the classic Faster R-CNN model and successfully applied it to impact crater detection using the THEMIS IR data, achieving a precision of 84.50% and a recall of 86.62%. Lagain et al. [38] trained YOLOv3 on High-Resolution Imaging Science Experiment (HiRISE) [39] images to detect small craters, later applying the model to Context Camera (CTX) [40] mosaics to focus on impact crater formation timing. Lagain et al. [41] further expanded this work by compiling a global database of approximately 90 million craters from CTX mosaics, facilitating secondary crater field analysis and meteorite launch site identification. Yang and Cai [42] proposed a high-resolution feature pyramid network to address the issue of imbalanced training samples for small craters, achieving a precision of 88.61% and recall of 83.23% on the Mars day crater detection dataset [42].

These state-of-the-art (SOTA) approaches demonstrate impressive accuracy in recognizing craters, proving that deep learning can significantly enhance crater extraction accuracy. However, most of these methods are not open-sourced. The few open-source algorithms available often require a deep understanding of machine learning concepts and intricate configuration steps, making them less accessible to researchers unfamiliar with computational setups. This high usage threshold poses a significant barrier to the broader scientific community's engagement with these powerful tools, limiting their utilization to planetary science scientists. Furthermore, existing methods tend to focus on specific celestial bodies and generally achieve high accuracy only on relatively simple, flat.

In response to the aforementioned problems, we developed "automatic crater detection (ACD) tool" as a solution to democratize and simplify the process of crater identification. The ACD tool adopts SOTA object detection algorithms tailored for each dataset to ensure precise and efficient crater detection on complex planetary surfaces, including the Moon, Mars, and Mercury. A standout feature of ACD tool is its user-friendly interface. Scientists from various backgrounds can easily perform crater identification with just a few clicks after downloading the tool.

This article is structured as follows. Section II provides a detailed description of the application of the deep learning model for crater detection, including dataset preparation, model introduction, and experimental setup. Section III presents the evaluation results of the crater detection model on the validation set. Section IV offers a comprehensive guide to using ACD tool. Finally, Section V summarizes the outcomes of this work.

II. Crater Detection Approach Based on Deep Learning Model

A. Overview of the Approach

The core of the ACD tool is a crater detection framework based on SOTA deep learning models. The development process began with the creation of a high-quality dataset consisting of craters from the Moon, Mars, and Mercury. This dataset was compiled using the you only look once (YOLO) model, a deep learning-based object detection framework. The model was then enhanced through multiple active learning cycles, using a training dataset that incorporated both publicly available crater databases [18], [43], [44] as well as targeted datasets manually annotated.

Extensive experiments were conducted on SOTA models to identify the most suitable model for crater detection on each planetary surface. The selected models were then fine-tuned and integrated into the ACD tool. Following crater extraction, geographical information for the detected craters was calculated. A flowchart detailing the entire crater detection process is presented in Fig. 1.

B. Training Sample Determination and Active Learning Process

In deep learning, the quality of training samples is a critical factor in determining the model's overall performance and accuracy. Representative data ensure that the training samples cover the various scenarios and conditions the model is expected to encounter. Therefore, this study has invested significant time and effort into curating high-quality training samples.

The first step in preparing the training samples is to acquire high-resolution satellite images of the Moon, Mars, and Mercury from publicly available databases from various space missions. For the lunar images, we utilized three different sources: 1) morphology mosaic [45], [46] from WAC aboard the LRO; 2) images from the LRO NAC; and 3) images from the SELENE Kaguya TC. For Mars, the images include: 1) THEMIS-IR mosaics; 2) images captured by the Mars Reconnaissance Orbiter's CTX; and 3) images from the HiRISE. The Mercury dataset comprises global mosaic images obtained using the Mercury Dual Imaging System (MDIS) aboard the MESSENGER spacecraft [47], [48].

The initial annotation of craters was sourced from publicly available online databases and supplemented with manually annotated data. For the Moon, the LROC WAC and SELENE TC images were annotated based on the studies by Wang

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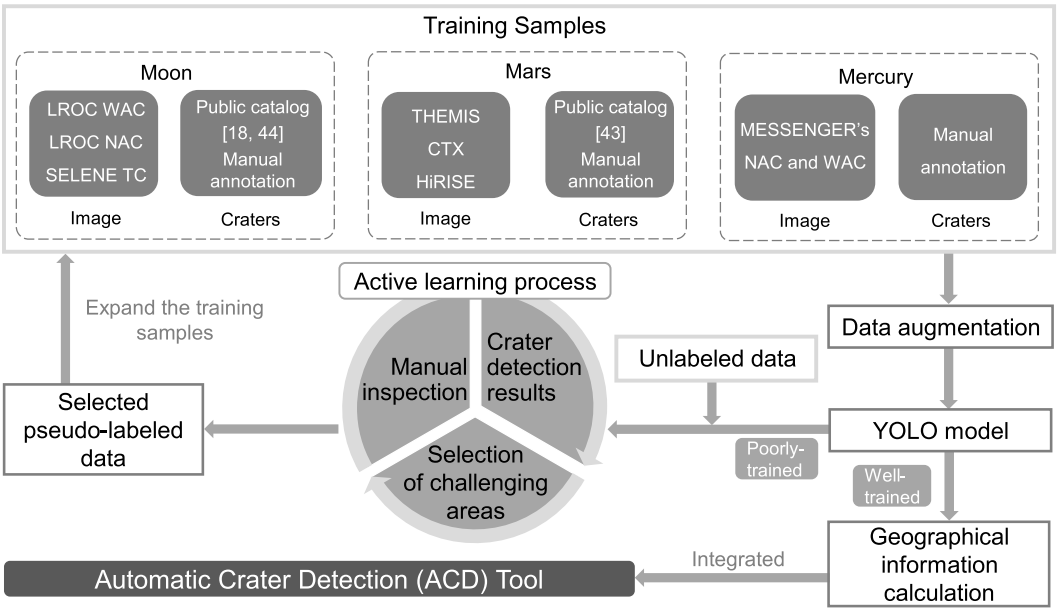

Fig. 1. Flowchart illustrating the whole process of the crater detection approach.

and Wu [18] and Robbins [44], respectively, while the LROC NAC images were manually annotated. For Mars, the crater annotation for the THEMIS-IR mosaics follows the work of Robbins and Hynek [43]. In addition, the CTX and HiRISE images received manual annotation. Regarding Mercury, the MDIS global mosaic was manually annotated and verified. Fig. 2 presents the partial annotation results of impact craters from various data sources, with each subfigure representing a specific image dataset.

After training the model with the initial samples, an active learning process was initiated to selectively expand the training dataset. We ran the model on a large dataset for crater detection and manually identified areas where the model performed poorly. These annotations were corrected and added to the training samples. The challenging areas, as examples shown in Fig. 3, often include poor lighting conditions, landforms with features resembling impact craters, and craters that have been partially modified. This targeted approach ensured that the model received the most relevant and challenging examples, accelerating its ability to distinguish between similar landforms and improving its overall performance and accuracy in a shorter time. This iterative process continued until the model's accuracy reached a satisfactory level, resulting in an expanded training sample set with high-quality annotated data.

Table I provides the detailed information about the final dataset used for training the crater detection approach.

To meet the input requirements of deep learning, ensure data diversity, and enhance model robustness, we further processed and optimized the training samples. First, to minimize the impact of different projection methods on crater shapes, all data were reprojected into the Mercator projection, which does not alter the circular shape of the craters. Second, to mitigate the effects of varying crater sizes, we excluded craters smaller than 6 pixels or larger than 320 pixels. Third, we divided the annotated planetary images into training, validation, and testing regions based on the height of the original images, maintaining an approximate ratio of 8:1:1 to prevent data leakage from overlapping tiles. After defining these regions, we segmented the annotated planetary images into 640 × 640 pixel tiles, with a 50% overlap both horizontally and vertically. For data augmentation, we focused on addressing the variations in illumination angles and directions caused by the different times of image capture. To adapt to various lighting conditions, each training sample was rotated eight times at 45° intervals across a 360° range, generating seven new annotated tiles. It is worth noting that actual model performance showed a decrease when using data augmentation methods other than rotation, such as scaling image size, mosaic [49], mix-up [50], and copy-paste [51]. Therefore, we only used rotation for data augmentation.

C. SOTA Models

To identify the most suitable model for crater detection on each planetary surface, experiments were conducted on SOTA models. Within the field of computer vision, three SOTA model families—R-CNN-based, YOLO-based, and Transformer-based architectures—each offer unique advantages and limitations. The selection of an optimal model architecture depends on the dataset's specific characteristics and the tradeoff between detection accuracy and inference efficiency.

R-CNN models, such as Faster R-CNN [24] and Cascade R-CNN [52], follow a two-stage pipeline: first generating candidate regions and then performing classification and bounding box regression. Although these multistep processes achieve high accuracy, they often result in slower inference speeds. In contrast, YOLO-based models—including YOLOv5 [32], YOLOv8 [53], YOLOv9 [54], YOLOv10 [55], and

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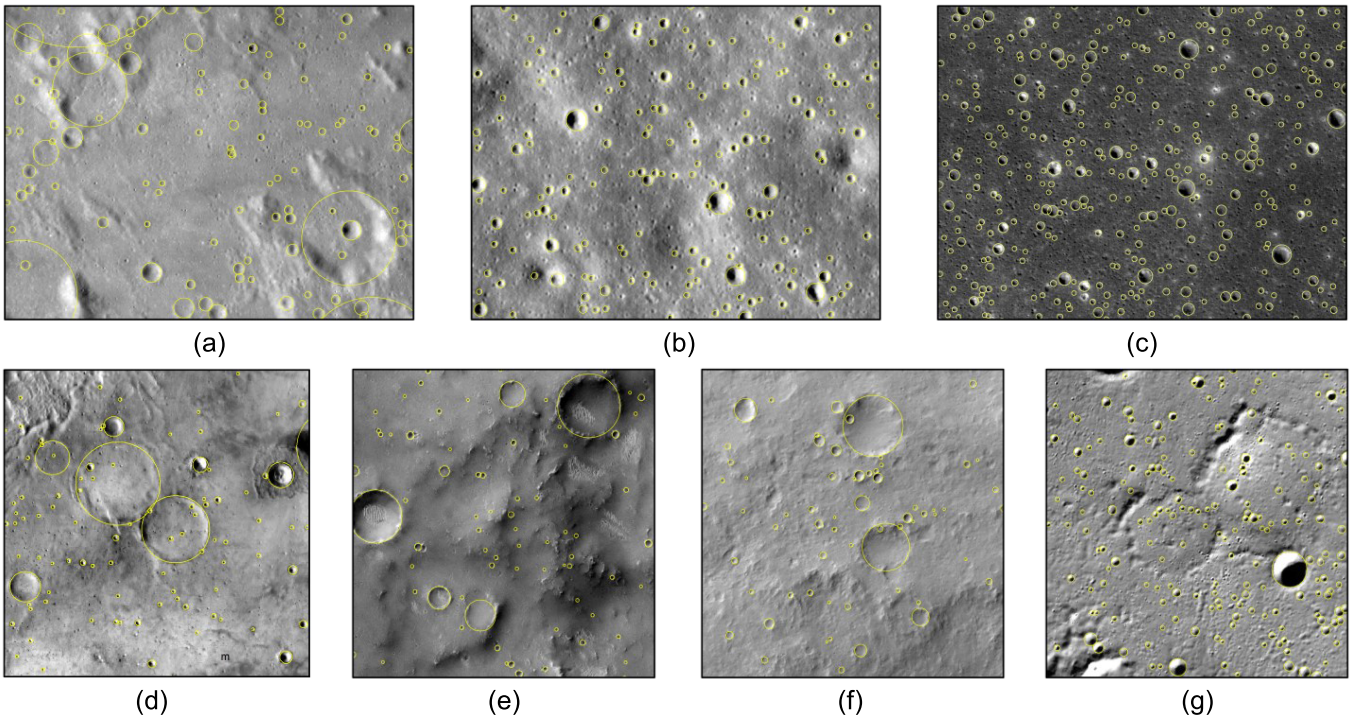

Fig. 2. Examples of crater annotations for different data sources. (a) LROC WAC. (b) LROC NAC. (c) SELENE TC. (d) THEMIS-IR. (e) CTX. (f) HiRISE. (g) MDIS.

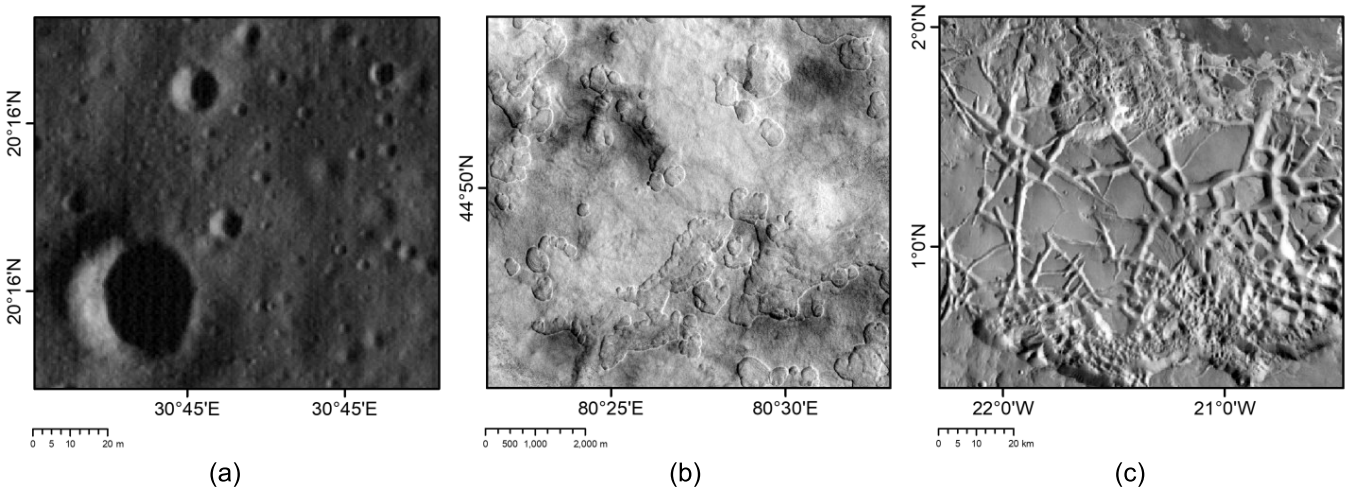

Fig. 3. Examples of challenging landforms for detection. (a) Regions with high solar incidence angles on the Moon, as seen in LROC NAC images. (b) Scalloped depressions on Mars, where structures with circular depressions can easily be mistaken for impact craters due to their similar appearance from CTX images. (c) Impact craters on Mars that have been modified by intricate cracks formed later, as shown in THEMIS-IR images.

YOLO11 [56]—use a single-stage framework where objects are detected in a single forward pass. This approach predicts bounding boxes and class probabilities simultaneously, enabling faster detection compared to other CNN architectures [30].

Transformer-based architectures, such as DINO [57] and DDQ-DETR [58], simplify the detection pipeline by treating object detection as a set prediction problem. By avoiding anchors, proposals, and postprocessing operations, these models introduce a more streamlined approach. Despite their higher computational demands, Transformers often achieve comparable or superior results on large-scale datasets.

Each of these model families—R-CNN, YOLO, and Transformer—strikes a distinct balance between accuracy and inference speed. In this study, we reproduce several SOTA architectures using the crater datasets (listed in Table I) and evaluate their performance in terms of precision and inference time.

D. YOLO Network

The crater detection approach is based on the YOLO model, an SOTA deep learning algorithm designed for real-time object detection, which processes images in a single pass to identify and classify objects efficiently and accurately.

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TABLE I
DATASETS USED FOR TRAINING THE CRATER DETECTION MODEL

Planets	Dataset	Resolution (m/pixel)	Source of Annotations	Crater Diameter Range (m)	Crater Number
Moon	LROC WAC	150	Robbins [44]	1,000~46,692	186,302
	LROC NAC	1.1	Manual annotation	7~307	34,012
	SELENE TC	6.2	Wang and Wu [18] + Manual annotation	120~1,949	54,563
Mars	THEMIS-IR	150	Robbins and Hynek [43]	994~47,322	121,517
	CTX	5	Manual annotation	21~2,467	6,678
	HiRISE	0.5~1	Manual annotation	2~144	5,978
Mercury	MDIS	166	Manual annotation	996~26,347	17,749

YOLOv8 [53], released by Ultralytics on January 10, 2023, represents a significant update in the YOLO series. YOLOv8 introduces advanced neural network architectures and training techniques to further enhance generalization across different datasets while maintaining high performance. The key improvements in YOLOv8 focus on feature fusion, detection head design, and loss function optimization. Specifically, the incorporation of a more gradient-rich C2f structure (C2f means a faster implementation of the cross-stage partial network [59] bottleneck with two convolutions) in the backbone and neck sections enhances feature representation. Additionally, the anchor-free design eliminates dependence on predefined anchor boxes [60], simplifying model design and hyperparameter tuning. The use of a decoupled head structure separates the classification and detection heads, optimizing the features needed for both tasks. The classification loss employs binary cross-entropy loss [61], while the bounding box regression loss uses complete intersection over union (IoU) loss [62] and distribution focal loss [63], collectively improving detection accuracy in complex scenarios.

Following the release of YOLOv8, the YOLOv9 [53] series introduces several architectural advancements. Notably, it incorporates the programmable gradient information (PGI) technique, which addresses the challenges of data loss during feature extraction in deep networks. By mitigating the information bottleneck, PGI preserves and effectively leverages gradient information, thereby optimizing the learning process. Additionally, YOLOv9 introduces the generalized efficient layer aggregation network (GELAN), a lightweight architecture that enhances parameter utilization. The integration of PGI and GELAN makes YOLOv9 particularly suitable for both lightweight and large-scale object detection tasks.

Building on YOLOv8, YOLO11 [56] prioritizes computational efficiency and performance. A key enhancement in YOLO11 is the replacement of the C2f block in YOLOv8 with the C3k2 block, which utilizes two smaller convolutions instead of a single large convolution, thus improving computational efficiency. YOLO11 optimizes feature extraction in the backbone and feature fusion in the neck section through the improved C3k2 block, resulting in faster inference speeds and enhanced overall accuracy.

These innovations make the YOLO family particularly effective in applications requiring fast and reliable object detection, such as crater detection for planetary surface dating [64], planetary surface navigation algorithms [65], and planetary surface navigation simulation data generation [66]. We utilized the YOLO-P2 configuration, which is particularly well-suited for the identification of small impact craters. The term "P2" refers to the use of specific pyramid levels of features. This approach emphasizes a balance between speed and accuracy, which is particularly beneficial for detecting smaller objects or fine-grained details. The structure of the YOLO-P2 model is illustrated in Fig. 4. The backbone is responsible for extracting features from the input image, transforming it into a feature representation rich in semantic information. The neck acts as an intermediate layer, fusing features from the backbone to enhance model performance. The head, the final layer of the model, includes both the bounding box regressor and the classifier, which is responsible for the ultimate object detection tasks.

E. Implementation

The training process consisted of multiple stages. Initially, SOTA models were trained on individual planetary crater datasets, and the most suitable model was selected for further fine-tuning. This optimized model was then integrated into the ACD tool. The final model was trained for 50 epochs using the SGD optimizer, with an initial learning rate of 0.01 and a momentum factor of 0.9.

Deep learning model training and testing using the crater datasets were conducted on a workstation configured with an Intel[1] Xeon[1] Gold 6230R CPU, an NVIDIA GeForce RTX 4090 GPU, Ubuntu 22.04 LTS operating system, PyTorch version 2.2.2, and CUDA version 12.1.

III. EVALUATIONS ON THE MOON, MARS, AND MERCURY

A. Evaluation Criteria

Extensive experiments were conducted on SOTA models to identify the most suitable model for crater detection on each planetary surface. To ensure a comprehensive evaluation, the following metrics were used: average precision at IoU > 0.5 (AP50), mean average precision over IoU thresholds ranging from 0.5 to 0.9 (AP50:90), frames per second

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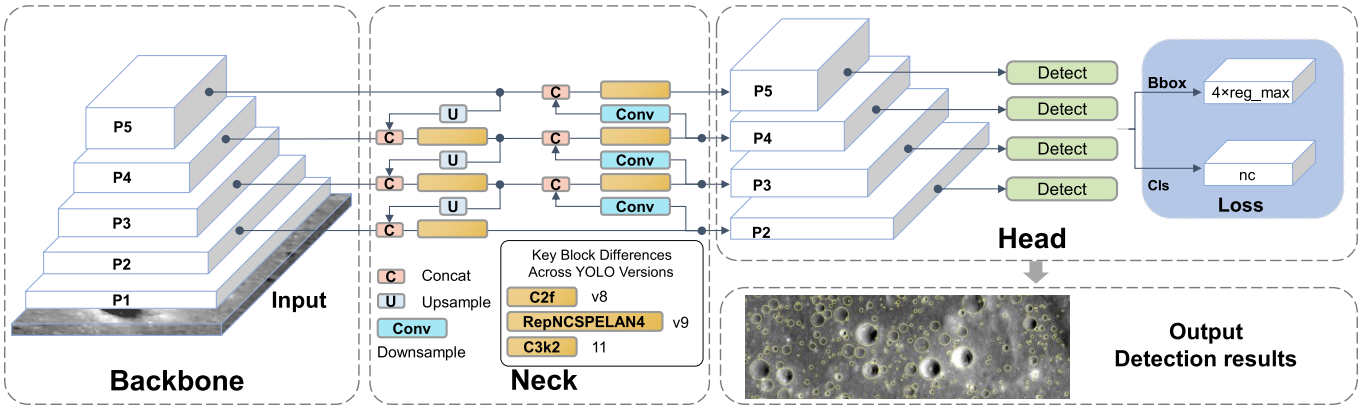

Fig. 4. Overview diagram of the impact crater detection model based on YOLO-P2.

(FPS), and the number of model parameters. These metrics provided a balanced view of model accuracy, efficiency, and computational demands, enabling the selection of a model that best fits the requirements of the ACD tool. The selected models were then fine-tuned and integrated into the ACD tool for further validation.

After the selection of the optimal model for crater detection based on the aforementioned metrics, we conducted extensive testing on different planetary datasets to validate the model's performance and ensure its reliability. Here are the evaluation results. Six test datasets were selected representing the general characteristics of the planetary surface on the Moon, Mars, and Mercury. Both the detection and evaluation areas contain diverse sizes and types of craters. The manually digitized crater catalog from the images served as the ground truth for evaluation. We focused on craters with diameters over 10 pixels, the minimum size needed to accurately capture crater shapes, which is a common threshold in crater detection algorithms [67], [68], [69].

The ACD tool's performance is evaluated using standard metrics called precision, recall, and $F1$-score

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{1}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{2}$$

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{3}$$

where true positive (TP) is the count of accurately identified crater detections (i.e., the detected craters are actual craters), with TP detections determined by an IoU $>$ 0.5. False negative (FN) represents missed detections (i.e., undetected craters), and false positive (FP) is the number of incorrect detections (i.e., detection of noncrater other objects as craters). To evaluate the accuracy of crater diameter extraction, we calculated the crater diameter error estimation (CDEE) as a percentage difference to evaluate the ACD tool's prediction accuracy for crater diameters. The CDEE was computed using the following formula [29]:

$$\text{CDEE} = \frac{D_{\text{TP}} - D_{\text{GT}}}{D_{\text{TP}}} \times 100 \tag{4}$$

where D_{TP} is the diameter of a detected crater classified as a TP and D_{GT} represents the diameter of the manually annotated ground truth impact crater matching the TP.

B. Performance of SOTA Models With Crater Dataset

The state-of-the-art models described in Section II-C are evaluated using the dataset presented in Table I for training, validation, and testing. Performance is evaluated using key metrics, including AP50 and AP50:95 for detection accuracy and FPS for detection speed. The results are presented in Table II.

Traditional models, such as Faster R-CNN and Cascade R-CNN, performed poorly on the Moon dataset, with AP50:95 scores of 27.2% and 27.6%, respectively. Due to their low accuracy, these models are unsuitable for crater detection tasks.

In contrast, the YOLO family of models demonstrated strong performance. We tested three different model sizes: -N represents the nano version, -T represents the tiny version, and -S denotes the small version. YOLOv5 and YOLOv8 achieved excellent results on the Moon and Mercury datasets but underperformed on Mars. YOLOv9-S demonstrated robust performance across all datasets, with the best results on the Moon (52.5%) and Mars (42.0%), and outperformed most models on Mercury with a score of 63.3%. It also outperformed the YOLOv10 series in terms of AP50:95 scores. Among the YOLO11 models, YOLO11-N excelled in balancing speed and accuracy, achieving AP50:95 scores of 64.2% for Mercury and 52.0% for the Moon. However, its performance on Mars was suboptimal.

The DDQ-DETR, a Transformer-based model, demonstrated competitive accuracy, with AP50:95 scores of 51.9% for the Moon and 67.4% for Mercury. However, its inference speed of 24.1 FPS was notably slower than the YOLO models, limiting its suitability for processing large-scale planetary remote sensing data. In comparison, DINO demonstrated poor performance in both accuracy and inference speed.

In summary, among SOTA models, YOLOv9-S excelled on both the Moon and Mars, while YOLO11-N outperformed other models on Mercury. Notably, YOLO11-S achieved a significant speed advantage on the Moon, with an inference speed of 147.3 FPS—twice that of YOLOv9-S—despite a modest 0.5% drop in AP50:95 compared to YOLOv9-S.

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TABLE II
OVERALL PERFORMANCE OF SOTA MODELS

	Datasets		Moon		Mars		Mercury	
Model	FPS	#Params(M)	AP50(%)	AP50:95(%)	AP50(%)	AP50:95(%)	AP50(%)	AP50:95(%)
Faster-RCNN	67.1	41.3	47.9	27.2	59.2	32.0	59.1	40.3
Cascade-RCNN	32.2	77.0	47.9	27.6	67.7	37.1	59.2	45.6
YOLOV5-N	176.7	2.5	75.8	49.6	70.2	40.4	76.4	58.1
YOLOV5-S	176.6	9.1	76.6	50.8	59.9	31.7	80.9	61.0
YOLOV8-N	**197.5**	3.0	76.2	50.1	69.6	40.4	82.0	61.9
YOLOV8-S	191.8	11.1	76.2	50.6	60.1	32.7	77.7	60.9
YOLOV9-T	76.7	**2.0**	77.3	50.9	70.4	40.8	82.2	63.4
YOLOV9-S	**70.4**	**7.2**	**78.1**	**52.5**	**72.2**	**42.0**	81.7	63.3
YOLOV10-N	128.2	2.7	75.1	50.2	67.4	39.5	80.2	60.5
YOLOV10-S	124.6	8.0	76.4	51.7	60.4	33.1	82.7	63.2
YOLO11-N	**153.0**	**2.6**	77.4	51.1	69.5	39.8	**83.4**	**64.2**
YOLO11-S	**147.3**	**9.4**	**77.8**	**52.0**	58.5	32.3	81.8	63.1
DINO	30.0	47.5	57.6	20.5	55.1	25.2	63.3	47.9
DDQ-DETR	24.1	48.3	**80.5**	51.9	71.7	38.7	**86.2**	**67.4**

TABLE III
EVALUATION RESULTS FOR MOON

Datasets	Image ID	Resolution (m/pixel)	Terrain Type	Incidence angle (°)	Diameter (m)	Precision (%)	Recall (%)	F1 (%)
LROC NAC	M1228163681L	0.5	Highland	45	⩾3	86.1	99.2	**92.2**
					⩾6	85.8	99.2	92.0
	M1422204907L	0.9	Mare	50	⩾7	89.8	97.3	**93.4**
					⩾14	91.5	95.7	93.6
	M1189051253L	0.5	Highland	70	⩾3	84.2	99.5	**91.2**
					⩾6	87.8	99.6	93.3
	M1173521337L	1.5	Mare	75	⩾8	86.6	99.6	**92.7**
					⩾12	83.1	99.4	90.5
SELENE TC	TCO_MAP_02_N00E000S03E003SC	7.4	Highland		⩾70	92.5	96.6	**94.5**
					⩾140	92.8	96.2	94.4
Average						**87.8**	**98.4**	**92.8**

The incorporation of the P2 feature layer enhanced detection accuracy, with improvements in AP50 scores of 3.5%–3.6% while maintaining relatively modest reductions in FPS.

Consequently, YOLO11-S-P2, YOLOv9-S-P2, and YOLO11-N-P2 were selected for crater detection on the Moon, Mars, and Mercury and integrated into the ACD tool.

C. Evaluations on LROC NAC and SELENE TC for Moon

To conduct a more comprehensive assessment of the ACD tool's performance under diverse conditions on the lunar surface, we carried out evaluations employing four NAC images and one SELENE TC image. NAC images represent lunar mare and highland and span a range of illumination conditions, with incidence angles between 45° and 75° (see Table III for details).

The evaluation for the Moon is on LROC NAC (0.9 m/pixel) and SELENE TC (7.5 m/pixel) dataset, as shown in Figs. 5(a) and 6(a), respectively. Yellow circles indicate TP, correct detection; blue circles indicate FN, missed detection; and red circles indicate FP, false detection. Figs. 5(b) and 6(b) show that, for Moon, the diameter error of impact craters detected by ACD is mostly within 10% and centered around zero. We consider this error acceptable, as Robbins et al. [70] also concluded that manual mapping of impact craters shows variability in size measurements of up to ±15%.

It can be seen that the missed detections usually occur in difficult cases where craters are highly eroded or irregular in shape, while the incorrect detections are always affected by the crater-like rugged surface. The quantitative evaluation results of the lunar impact crater datasets are shown in

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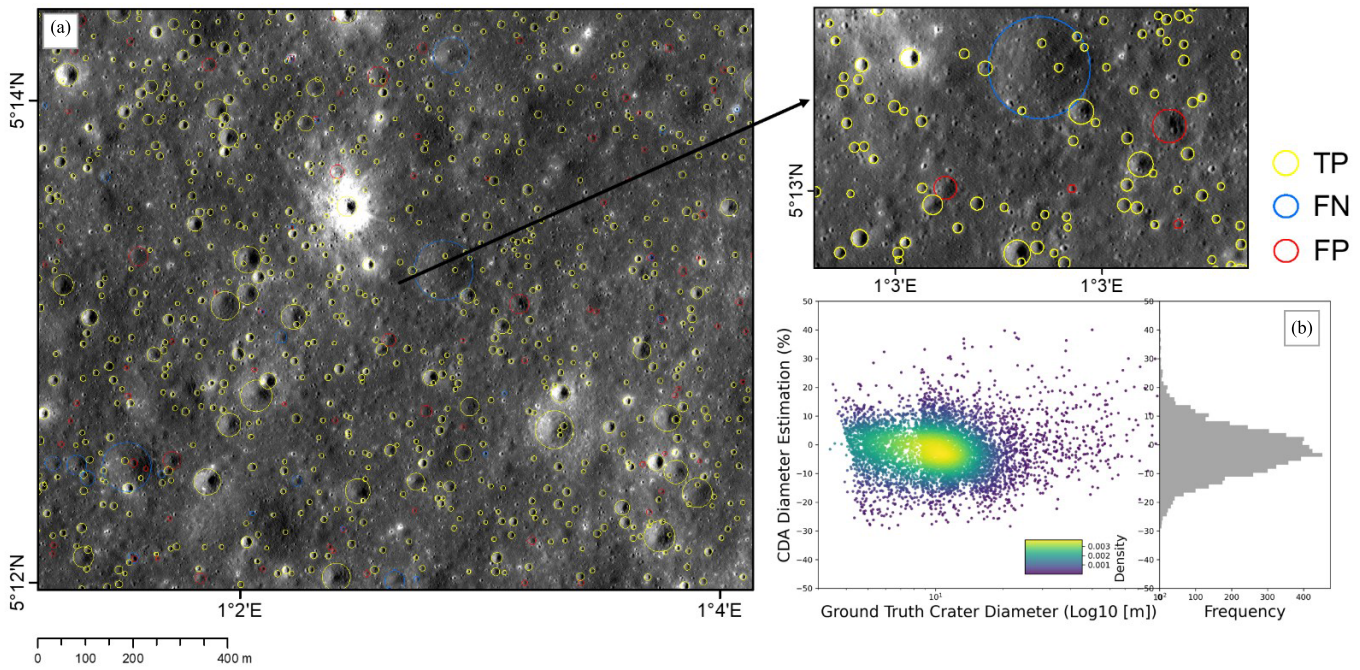

Fig. 5. (a) Evaluations on LROC NAC (image ID: M1422204907LE) for the Moon. Yellow circles indicate correct detection, blue is for missed detection, and red is for false detection. (b) Crater diameter estimation based on ground truth. Point color coding is computed using kernel density estimation (KDE), which visualizes the spatial density of points relative to each other; yellow indicates higher density, while blue indicates lower density.

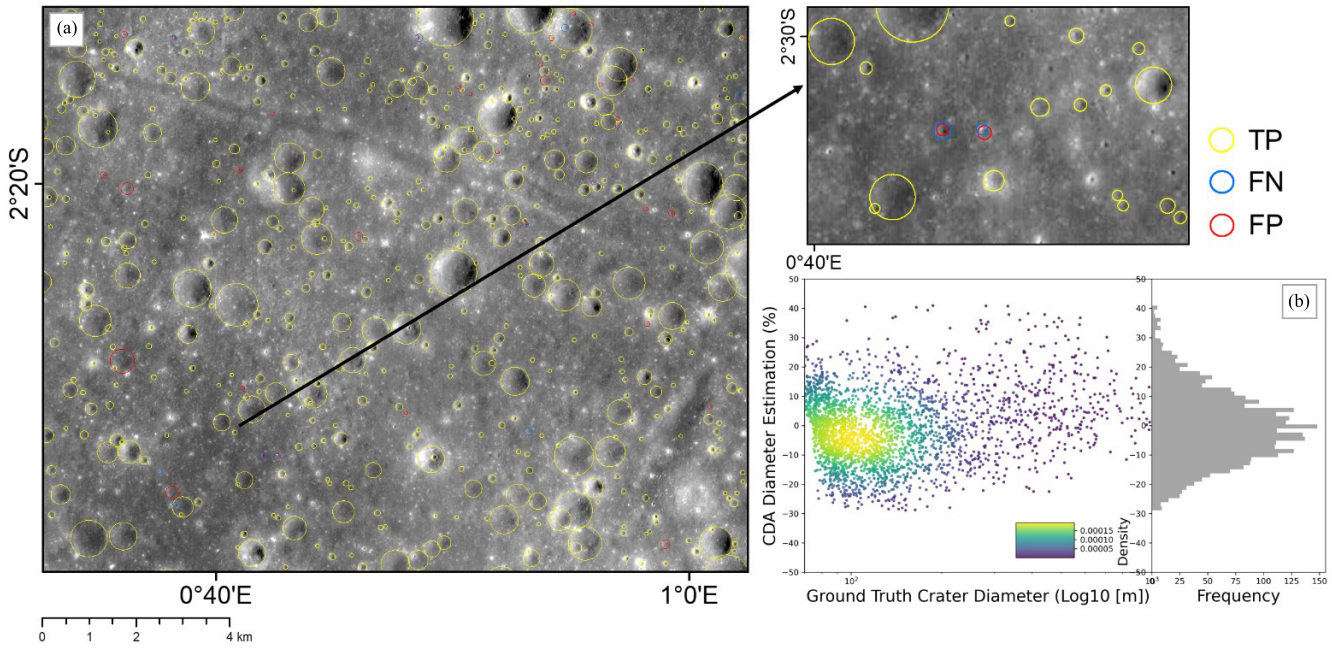

Fig. 6. (a) Evaluations on SELENE TC (image ID: TCO_MAP_02_N00E000S03E003SC) for the Moon. Yellow circles indicate correct detection, blue is for missed detection, and red is for false detection. (b) Crater diameter estimation based on ground truth. Point color coding is computed using KDE, which visualizes the spatial density of points relative to each other; yellow indicates higher density, while blue indicates lower density.

Table III, revealing that the average $F1$-score for Moon craters reached 92.8%.

D. Evaluations on THEMIS-IR, CTX, and HiRISE for Mars

To assess the performance of ACD on Mars, we conducted evaluations using seven CTX images, one THEMIS-IR, and one HiRISE image. CTX images represent six Terrain Types (see Table IV for details) [41].

The experiment conducted on the Mars data followed a similar process to that used for the Moon data. The evaluation for Mars was performed on datasets from THEMIS-IR (100 m/pixel), CTX (5 m/pixel), and HiRISE (0.25 m/pixel), as shown in Figs. 7(a), 8(a), and 9(a). According to the quantitative evaluation results presented in Table IV, the average $F1$-scores reached 91.7%, which is comparable to the performance on the Moon. For Mars, the diameter error

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TABLE IV
EVALUATION RESULTS FOR MARS

Datasets	Image ID	Resolution (m/pixel)	Terrain Type	Diameter (m)	Precision (%)	Recall (%)	F1 (%)
THEMIS -IR		100		⩾1,000	90.2	90.5	**90.4**
				⩾2,000	94.3	94.3	94.3
CTX	MurrayLab_CTX_V01 _E000_N-16	5	Highland	⩾50	93.2	96.5	**94.8**
				⩾100	96.2	98.8	97.5
	MurrayLab_CTX_V01 _E-100_N24	5	Volcanic	⩾50	87.8	96.1	**91.8**
				⩾100	84.3	98.8	91.0
	MurrayLab_CTX_V01 _E088_N12	5	Lowland	⩾50	93.9	86.9	**90.3**
				⩾75	96.1	95.5	95.8
	MurrayLab_CTX_V01 _E-144_N24	5	Apron	⩾50	95.9	90.4	**93.1**
				⩾100	98.0	95.1	96.6
	MurrayLab_CTX_V01 _E092_N08	5	Impact	⩾50	86.8	88.3	**87.6**
				⩾100	93.0	90.3	91.6
	MurrayLab_CTX_V01 _E020_N48	5	High latitude	⩾50	87.3	93.8	**90.4**
				⩾100	90.0	95.0	92.4
HiRISE	ESP_028784_1800_R ED	0.3	Highland	⩾2.5	94.9	95.5	**95.2**
				⩾5	94.0	93.7	93.9
Average					**91.2**	**92.2**	**91.7**

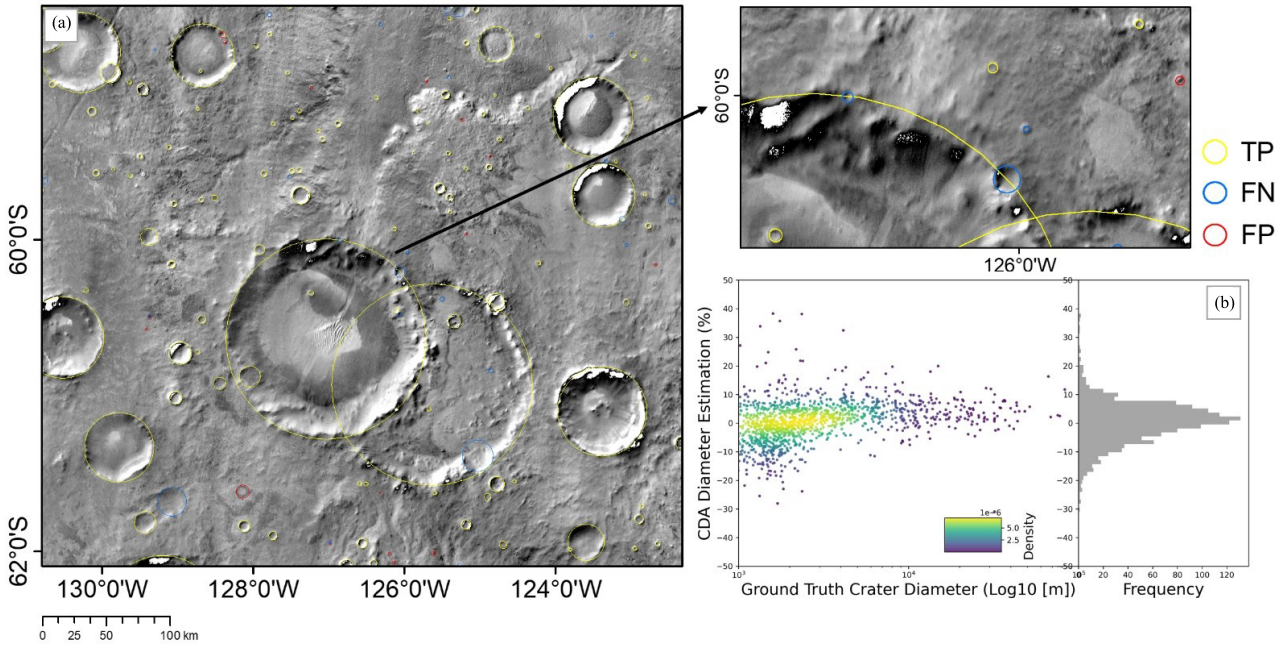

Fig. 7. (a) Evaluations on THEMIS-IR (image ID: Mars_MO_THEMIS-IR-Day_mosaic_global_100 m_v12; location: 61°S, 129°W) for Mars. Yellow circles indicate correct detection, blue is for missed detection, and red is for false detection. (b) Crater diameter estimation based on ground truth. Point color coding is computed using KDE, which visualizes the spatial density of points relative to each other; yellow indicates higher density, while blue indicates lower density.

of impact craters detected by ACD is mostly within 10% and centers around zero [Figs. 7(b), 8(b), and 9(b)], similar to the Moon.

E. Evaluations on MDIS for Mercury

To assess the performance of ACD on Mercury, we conducted evaluations using three MDIS NAC images and one section of an MDIS mosaic image. The MDIS NAC images span a range of illumination conditions, with incidence angles between 45° and 65° (see Table V for details). The experiment conducted on the Mercury data followed a similar process as that on the Moon and Mars data. The evaluation for Mercury is performed on the MDIS dataset (250 m/pixel), as illustrated in Fig. 10(a). Missed detections of impact craters primarily

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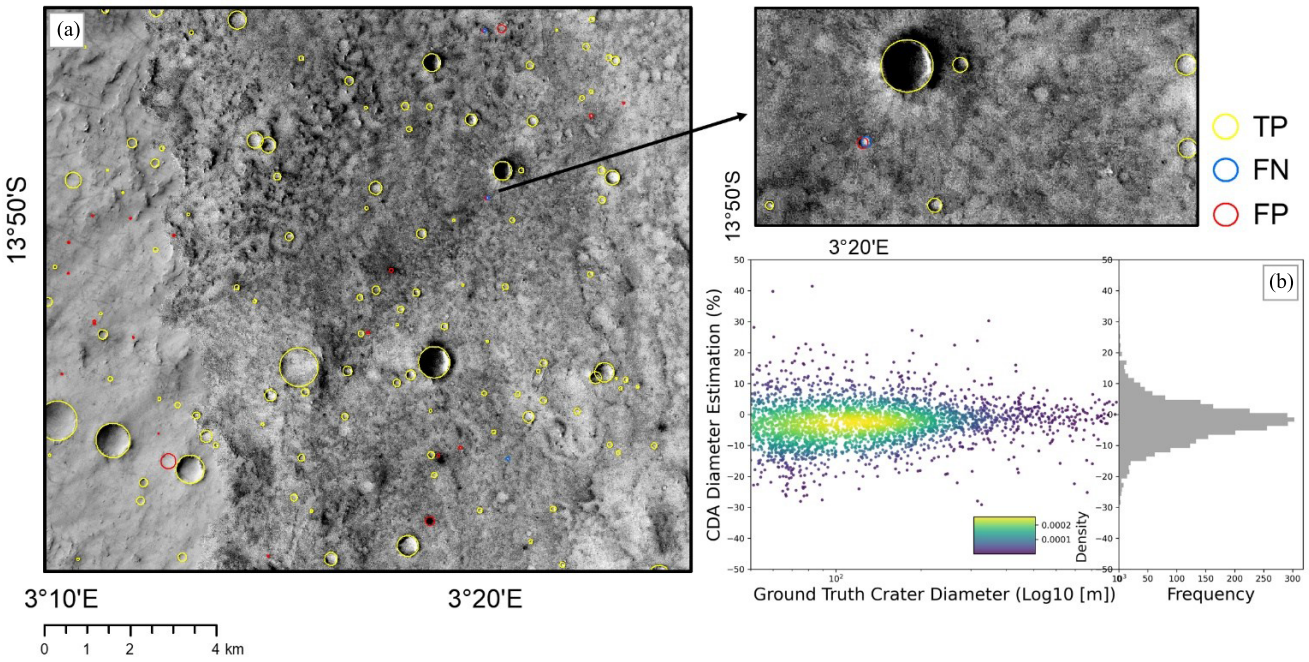

Fig. 8. (a) Evaluations on CTX (image ID: MurrayLab_CTX_V01_E000_N-16) for Mars. Yellow circles indicate correct detection, blue is for missed detection, and red is for false detection. (b) Crater diameter estimation based on ground truth. Point color coding is computed using KDE, which visualizes the spatial density of points relative to each other; yellow indicates higher density, while blue indicates lower density.

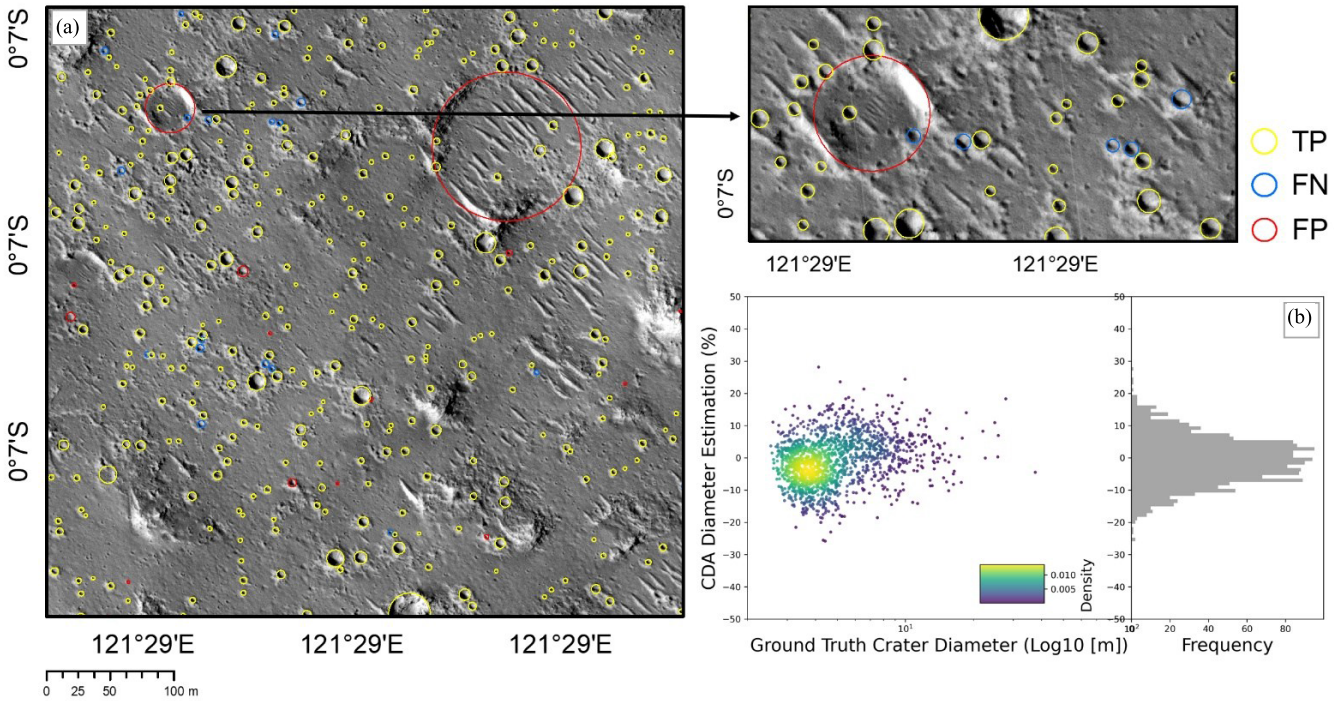

Fig. 9. (a) Evaluations on HiRISE (image ID: esp_028784_1800_red) for Mars. Yellow circles indicate correct detection, blue is for missed detection, and red is for false detection. (b) Crater diameter estimation based on ground truth. Point color coding is computed using KDE, which visualizes the spatial density of points relative to each other; yellow indicates higher density, while blue indicates lower density.

occur with those that are highly degraded. In this test image of Mercury, the craters are more densely packed, with many being adjacent or overlapping. Despite this complexity, the $F1$-score still reached 92.2%, as detailed in Table V. Fig. 10(b) shows that, for Mercury, the diameter error of impact craters detected by ACD is an overestimation of approximately 10%.

IV. User Interface and Accessibility of ACD Tool

The well-trained crater detection model was integrated into the ACD tool, allowing users to perform ACD on new planetary surface images. The ACD tool runs on a Windows system with an intuitive user interface (as shown in Fig. 11). ACD tool ensures easy utilization with only a double click of

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TABLE V
EVALUATION RESULTS FOR MERCURY

Datasets	Resolution (m/pixel)	Diameter (m)	Precision (%)	Recall (%)	F1 (%)
MDIS	250	⩾2,500	97.2	87.6	**92.2**
		⩾4,000	98.5	85.7	91.6

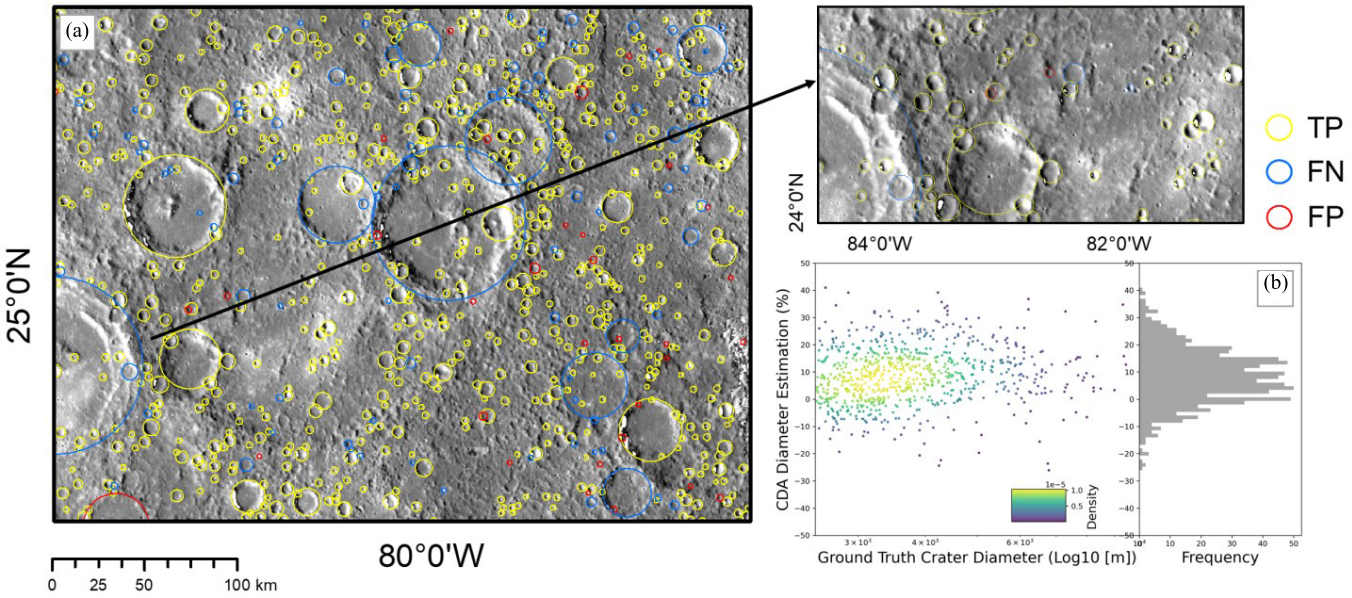

Fig. 10. (a) Evaluations on MDIS (image name: Mercury_MESSENGER_mosaic_global_250 m_2013; location: 25.5°N, 81°W) for Mercury. Yellow circles indicate correct detection, blue is for missed detection, and red is for false detection. (b) Crater diameter estimation based on ground truth. Point color coding is computed using KDE, which visualizes the spatial density of points relative to each other; yellow indicates higher density, while blue indicates lower density.

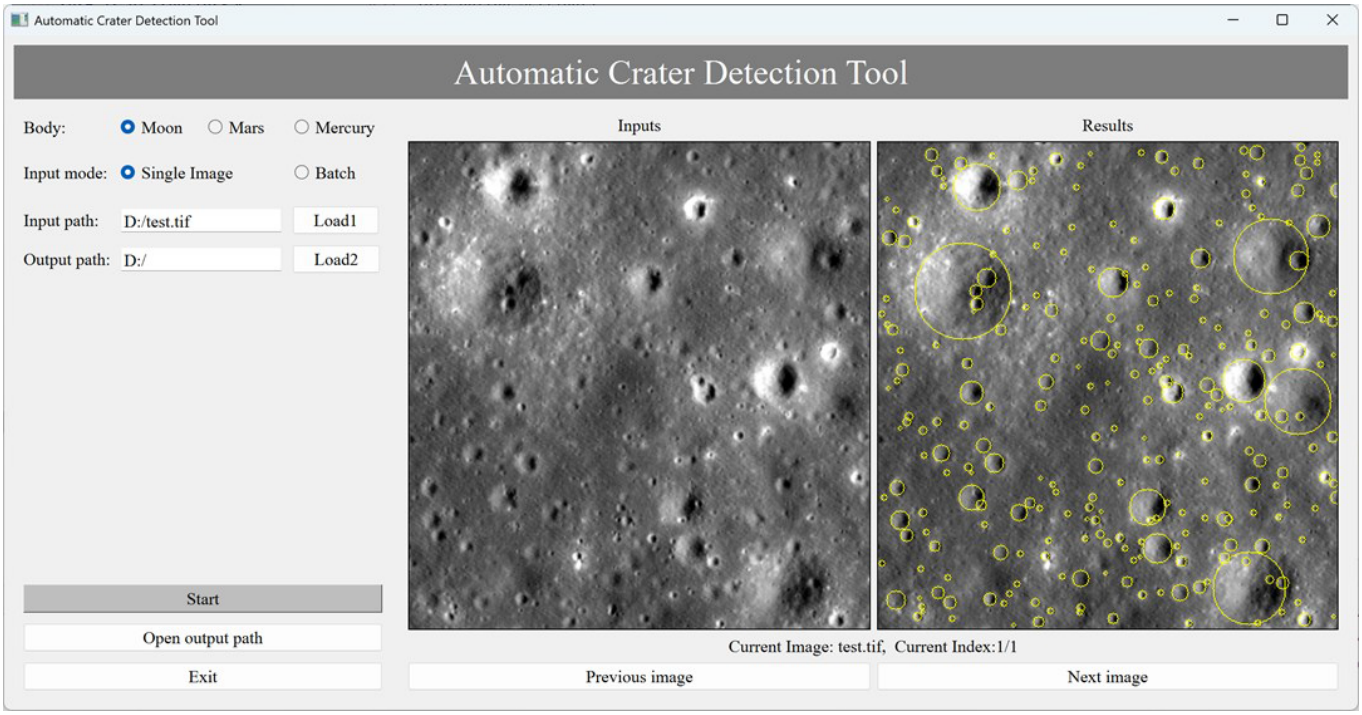

Fig. 11. Interface of the ACD tool. The left side serves as the input and control panel, while the right side functions as the display area.

the exe file and does not require additional system environment configurations. The ACD tool is available in both CPU and GPU versions, allowing users to choose according to their needs. Please note that the tool may take some time (5

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s or more, depending on the computer's performance) to initialize.

The interface of the ACD tool is divided into two parts: the left side serves as the input and control panel, while the right side functions as the display area. Users can select among three celestial bodies—Mars, Moon, or Mercury—by clicking on the respective options. The software offers two input modes. In the "single image" mode, users directly enter the image paths. In the "batch" mode, the program processes all TIFF files within a specified folder. Input images should be GeoTIFF files. The GeoTIFF files can have any projected coordinates, and they will all be automatically converted to Mercator projection by the ACD tool to ensure circular shapes for impact craters.

The ACD process begins with a click of the "Start" button. Upon completion, the results are saved in the specified output path in both shapefile and TXT file formats. The attribute table of the shapefile includes the crater's central latitude and longitude ("Lon" and "Lat") and the geodesic diameter in meters ("Dia"). The tool accounts for and corrects projection distortion through built-in computations. As the craters are extracted in the Mercator projection, the corresponding information is provided to better correspond with the images: "X_Merc" and "Y_Merc" for the x- and y-coordinates in Mercator and "D_Merc" for the distorted diameter measured in the Mercator projection. The confidence score ("Score"), ranging from 0 to 1, is also provided; higher values indicate greater confidence in the crater detection results. Additionally, a TXT file containing information identical to the shapefile attributes is generated. If an error occurs during the detection process, the names of the erroneous images and the reasons are recorded in a log file.

The interface on the right provides a preview image displaying the crater detection results. In the batch mode, the "Previous Image" and "Next Image" buttons allow sequential preview of all images and detection results.

V. Conclusion and Discussion

In response to the challenges associated with the usability of crater identification methods, this study introduces the ACD tool—a user-friendly software designed to facilitate the detection of impact craters on planetary bodies such as the Moon, Mars, and Mercury.

The core of the ACD tool is a crater detection method based on the YOLO deep learning algorithm, which was selected through experimental parameter comparisons. The utilization of active learning techniques allows the tool to adapt and improve with targeted data, enhancing its performance in challenging scenarios such as craters under low illumination angles, complex surfaces with crater-like features, or craters that have been modified and degraded. The tool can detect craters with a minimum diameter of 10 pixels. Evaluations on standard image datasets from the Moon, Mars, and Mercury yielded $F1$-scores of 92.8%, 91.7%, and 92.2%, respectively.

ACD tool is wrapped with an interactive user interface that ensures easy utilization, allowing users to generate ready-to-use output data with a simple click. Users can perform batch crater detection by directly inputting the folder containing the images. The output includes a shapefile and a text document containing the latitude, longitude, diameter, and confidence scores of the craters. This information can be directly used for subsequent scientific analyses, such as surface age estimation.

However, the ACD tool does have several limitations.

1) The ACD tool inevitably has errors in recognition and missed detections, so it is necessary to supplement with manual checks before using the crater results.
2) The diameter error of impact craters is centered around zero for the Moon and Mars but shows an overestimation of approximately 10% for Mercury.
3) Misidentification mainly stems from distinguishing between degraded craters with unclear edges and topographical undulations, which is also unclear for manual identification. Therefore, the manual identification results used for evaluation are not guaranteed to be completely accurate. Another limitation lies in the software itself. The exe version of the interfaced ACD tool may occasionally crash due to compatibility issues between the packaged operating environment and different computers. For a more stable operation, users are advised to utilize our open-source code and configure a local deep learning environment.

Future work will focus on expanding the capabilities of the ACD tool to include additional celestial bodies and refining the detection algorithms to handle increasingly complex datasets.

We will continuously collect user feedback, and new versions of the software will be periodically updated and released. The software can be downloaded via the following link: https://www.scidb.cn/s/yemuYb.

Acknowledgment

The authors express their deep gratitude to Robbins for generously making the Moon and Mars crater datasets publicly available. They also extend their thanks to the entire teams of the Lunar Reconnaissance Orbiter (LRO), Kaguya, MO, and MRO missions, including the principal investigators, staff, and students involved in data acquisition and analysis, for their exemplary leadership and dedication.

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